PRACTICAL 7

PHP Form Handling, Validation, Sanitization & File Storage

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1. Aim

To create a PHP registration/contact form that submits data using POST, validates and sanitizes user input, stores submitted data in a CSV or JSON file, and displays success and validation-error messages.

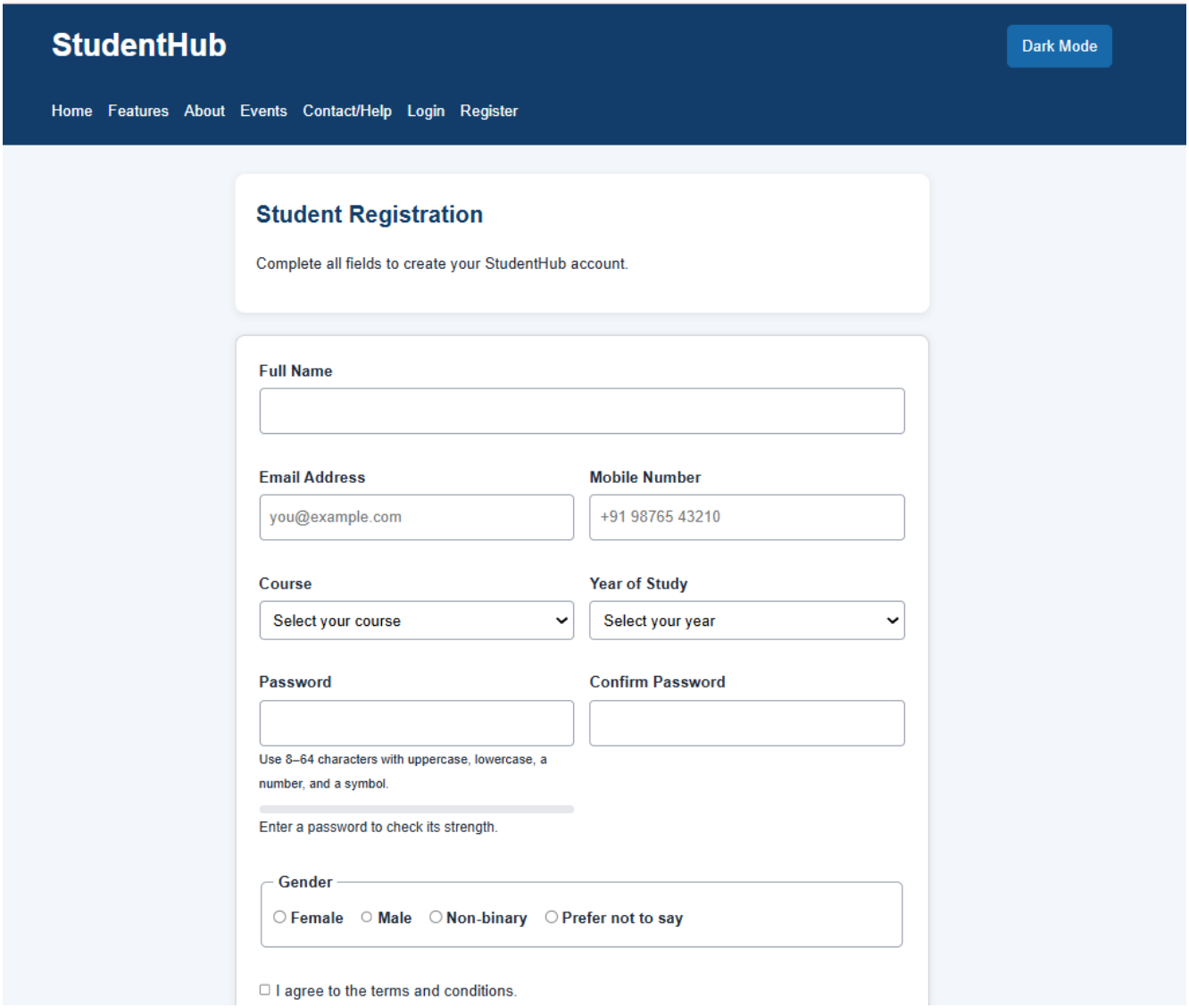
2. Main Submission Requirements

- PHP Form – Create a registration/contact form.
- POST Method – Submit form data using method="POST".
- Validation – Check for empty or invalid inputs.
- Sanitization – Clean user input before processing or storing it.
- File Storage – Store submitted data in a CSV or JSON file.
- Success Message – Show successful submission.
- Error Message – Show validation errors.

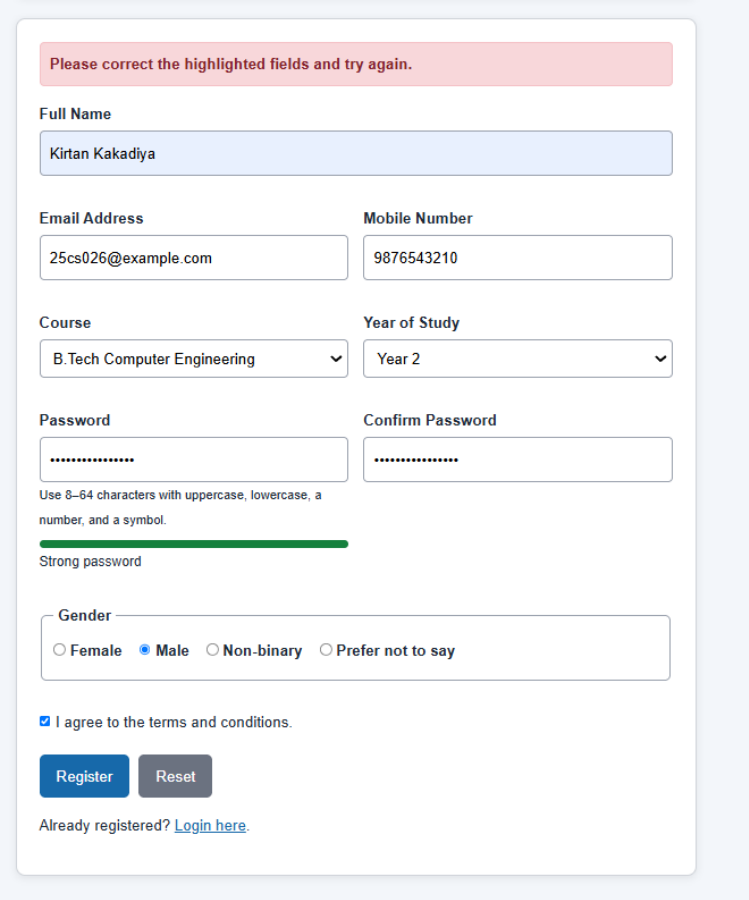
3. Required Screenshots

Paste a clear screenshot into each placeholder. Keep the relevant form state or message readable.

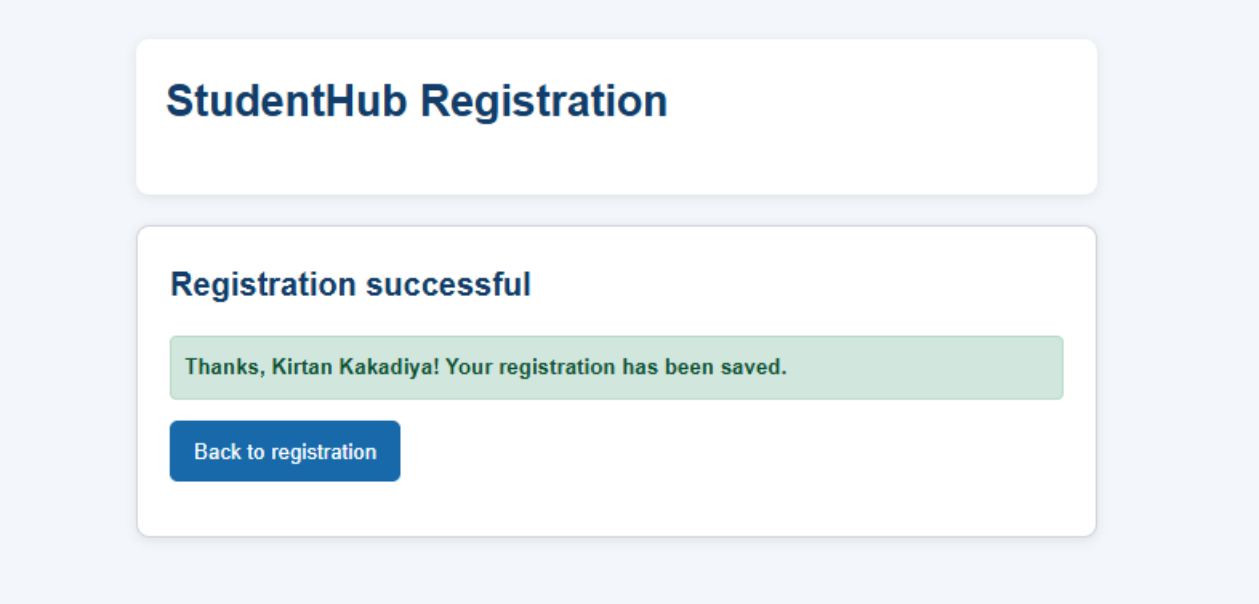
3.1 Form UI


StudentHub
Dark Mode
Home Features About Events Contact/Help Login Register
Student Registration
Complete all fields to create your StudentHub account.
Full Name
Email Address
you@example.com
Mobile Number
+91 98765 43210
Course
Select your course
Year of Study
Select your year
Password
Confirm Password
Use 8–64 characters with uppercase, lowercase, a number, and a symbol.
Enter a password to check its strength.
Gender
Female Male Non-binary Prefer not to say
I agree to the terms and conditions.

3.2 Successful Submission



>>>then,



3.3 Validation / Error Message

Empty field Register>>

Please correct the highlighted fields and try again.

Full Name

Enter a name with 2–60 letters. Spaces, apostrophes, periods, and hyphens are allowed.

Email Address

you@example.com

Enter a valid email address, such as student@example.com.

Mobile Number

+91 98765 43210

Enter a valid mobile number with 8–15 digits. A leading + and separators are allowed.

Course

Select your course

Select your course.

Year of Study

Select your year

Select your year of study.

Password

Use 8–64 characters with uppercase, lowercase, a number, and a symbol.

Enter a password to check its strength.

Use 8–64 characters with uppercase and lowercase letters, a number, and a symbol. Spaces are not allowed.

Confirm Password

Confirm your password.

Gender

○ Female ○ Male ○ Non-binary ○ Prefer not to say

Select a gender option, or choose “Prefer not to say”.

☐ I agree to the terms and conditions.

You must accept the terms and conditions to register.

3.4 CSV / JSON Stored Data

registrations.json

C: > xampp > htdocs > studentHUB > data > {} registrations.json > ...

```json
[
    {
        "name": "Kirtan Kakadiya",
        "email": "25cs026@example.com",
        "mobile": "9876543210",
        "course": "B.Tech Computer Engineering",
        "year": "Year 2",
        "gender": "Male",
        "passwordHash": "$2y$10$dd4gGz2q4rx3MMY3TZrdd.8rUTAVXg7CcxVh0agj1b0h4x.FoxNOW",
        "submittedAt": "2026-10-28T07:04:45+01:00"
    }
]
```

4. Recommended Technical Evidence

These code screenshots are recommended if the instructor checks how the requirements were implemented.

4.1 HTML Form and POST Method

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Register - StudentHub</title>
    <link rel="stylesheet" href="../css/common.css" />
    <link rel="stylesheet" href="../css/grid.css" />
    <script src="../js/auth.js" defer></script>
    <script src="../js/ui.js" defer></script>
  </head>
  <body data-auth-page="register">
    <header class="site-header">
      <div class="container">
        <h1><a href="../home.html">StudentHub</a></h1>
        <div class="header-controls">
          <button type="button" class="theme-toggle" data-theme-toggle aria-
label="Toggle light and dark theme">Dark Mode</button>
          <button type="button" class="menu-toggle" aria-label="Toggle navigation
menu" aria-expanded="false" aria-controls="main-navigation">&#9776; Menu</button>
        </div>
        <nav id="main-navigation" class="main-nav" aria-label="Main Navigation">
          <a href="../home.html">Home</a>
          <a href="features.html">Features</a>
          <a href="about.html">About</a>
```

```
          <a href="events.html">Events</a>
          <a href="contact-help.html">Contact/Help</a>
          <a href="login.html">Login</a>
          <a href="register.html" aria-current="page">Register</a>
        </nav>
      </div>
    </header>

    <main class="auth-container">
      <section class="page-title">
        <h2>Student Registration</h2>
        <p>Complete all fields to create your StudentHub account.</p>
      </section>

      <section class="auth-card">
        <p id="registrationMessage" class="form-message" role="alert" aria-
live="assertive" tabindex="-1"></p>
        <form action="../process.php" method="POST" id="registrationForm"
class="register-grid" novalidate>
          <div class="form-group full-width">
            <label for="student-name">Full Name</label>
            <input id="student-name" name="name" type="text" autocomplete="name"
required minlength="2" maxlength="60" aria-describedby="student-name-error" />
            <p class="field-error" id="student-name-error"></p>
          </div>

          <div class="form-group">
            <label for="email">Email Address</label>
            <input id="email" name="email" type="email" autocomplete="email" required
maxlength="254" placeholder="you@example.com" aria-describedby="email-error" />
            <p class="field-error" id="email-error"></p>
          </div>

          <div class="form-group">
            <label for="mobile">Mobile Number</label>
            <input id="mobile" name="mobile" type="tel" autocomplete="tel"
inputmode="tel" required minlength="8" maxlength="20" placeholder="+91 98765 43210"
aria-describedby="mobile-error" />
            <p class="field-error" id="mobile-error"></p>
          </div>

          <div class="form-group">
            <label for="course">Course</label>
            <select id="course" name="course" required aria-describedby="course-
error">
              <option value="">Select your course</option>
              <option value="B.Tech Computer Engineering">B.Tech Computer
Engineering</option>
```

```
              <option value="B.Tech Information Technology">B.Tech Information
Technology</option>
              <option value="Bachelor of Computer Applications">Bachelor of Computer
Applications</option>
            </select>
            <p class="field-error" id="course-error"></p>
          </div>

          <div class="form-group">
            <label for="year">Year of Study</label>
            <select id="year" name="year" required aria-describedby="year-error">
              <option value="">Select your year</option>
              <option value="Year 1">Year 1</option>
              <option value="Year 2">Year 2</option>
              <option value="Year 3">Year 3</option>
              <option value="Year 4">Year 4</option>
            </select>
            <p class="field-error" id="year-error"></p>
          </div>

          <div class="form-group">
            <label for="password">Password</label>
            <input id="password" name="password" type="password" autocomplete="new-
password" required minlength="8" maxlength="64" aria-describedby="password-help
password-strength password-error" />
            <small id="password-help">Use 8–64 characters with uppercase, lowercase, a
number, and a symbol.</small>
            <div id="password-strength" class="password-strength" role="meter" aria-
label="Password strength" aria-valuemin="0" aria-valuemax="6" aria-valuenow="0">
              <span class="password-strength-bar"></span>
            </div>
            <p class="strength-label" id="password-strength-label" aria-
live="polite">Enter a password to check its strength.</p>
            <p class="field-error" id="password-error"></p>
          </div>

          <div class="form-group">
            <label for="confirm-password">Confirm Password</label>
            <input id="confirm-password" name="confirmPassword" type="password"
autocomplete="new-password" required minlength="8" maxlength="64" aria-
describedby="confirm-password-error" />
            <p class="field-error" id="confirm-password-error"></p>
          </div>

          <fieldset class="form-group choice-group full-width" id="gender-group" aria-
describedby="gender-error">
            <legend>Gender</legend>
            <div class="choice-options">
```

```
              <label><input type="radio" name="gender" value="Female" required />
Female</label>
              <label><input type="radio" name="gender" value="Male" /> Male</label>
              <label><input type="radio" name="gender" value="Non-binary" /> Non-
binary</label>
              <label><input type="radio" name="gender" value="Prefer not to say" />
Prefer not to say</label>
            </div>
            <p class="field-error" id="gender-error"></p>
          </fieldset>

          <div class="checkbox-group full-width">
            <input id="terms" name="terms" type="checkbox" required aria-
describedby="terms-error" />
            <label for="terms">I agree to the terms and conditions.</label>
            <p class="field-error" id="terms-error"></p>
          </div>

          <div class="form-actions full-width">
            <button type="submit">Register</button>
            <button type="reset">Reset</button>
          </div>
        </form>
        <p>Already registered? <a href="login.html">Login here</a>.</p>
      </section>
    </main>

    <footer class="site-footer"><p>&copy; 2026 StudentHub. All Rights
Reserved.</p></footer>
  </body>
</html>
```

4.2 PHP Code

```
<?php

function escapeHtml($value)
{
    return htmlspecialchars($value, ENT_QUOTES | ENT_SUBSTITUTE, 'UTF-8');
}

function showResult($title, $message, $errors = [], $statusCode = 200)
{
    http_response_code($statusCode);
    $safeTitle = escapeHtml($title);
```

```
    $safeMessage = escapeHtml($message);

    echo '<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>' . $safeTitle . ' - StudentHub</title>
  <link rel="stylesheet" href="css/common.css">
</head>
<body>
  <main class="auth-container">
    <section class="page-title"><h1>StudentHub Registration</h1></section>
    <section class="auth-card">
      <h2>' . $safeTitle . '</h2>
      <p class="form-message ' . ($statusCode >= 400 ? 'error-message' : 'success-
message') . '">' . $safeMessage . '</p>';

    if ($errors) {
        echo '<ul>';
        foreach ($errors as $error) {
            echo '<li>' . escapeHtml($error) . '</li>';
        }
        echo '</ul>';
    }

    echo '<p><a class="button" href="pages/register.html">Back to registration</a></p>
    </section>
  </main>
</body>
</html>';
}

function postValue($key)
{
    if (!isset($_POST[$key]) || !is_string($_POST[$key])) {
        return '';
    }

    return trim($_POST[$key]);
}

function saveRegistration($registration)
{
    $directory = __DIR__ . DIRECTORY_SEPARATOR . 'data';
    if (!is_dir($directory) && !mkdir($directory, 0775, true) && !is_dir($directory))
{
        return 'error';
```

```
    }

    $path = $directory . DIRECTORY_SEPARATOR . 'registrations.json';
    $file = fopen($path, 'c+');
    if ($file === false) {
        return 'error';
    }

    if (!flock($file, LOCK_EX)) {
        fclose($file);
        return 'error';
    }

    rewind($file);
    $contents = stream_get_contents($file);
    if ($contents === false) {
        flock($file, LOCK_UN);
        fclose($file);
        return 'error';
    }

    if (trim($contents) === '') {
        $registrations = [];
    } else {
        $decoded = json_decode($contents);
        if (json_last_error() !== JSON_ERROR_NONE || !is_array($decoded)) {
            flock($file, LOCK_UN);
            fclose($file);
            return 'error';
        }

        $registrations = json_decode($contents, true);
    }

    foreach ($registrations as $savedRegistration) {
        if (
            is_array($savedRegistration)
            && isset($savedRegistration['email'])
            && strtolower($savedRegistration['email']) ===
strtolower($registration['email'])
        ) {
            flock($file, LOCK_UN);
            fclose($file);
            return 'duplicate';
        }
    }

    $registrations[] = $registration;
```

```
    $json = json_encode($registrations, JSON_PRETTY_PRINT | JSON_UNESCAPED_UNICODE);
    if ($json === false) {
        flock($file, LOCK_UN);
        fclose($file);
        return 'error';
    }

    rewind($file);
    $output = $json . PHP_EOL;
    $bytesWritten = ftruncate($file, 0) ? fwrite($file, $output) : false;
    $written = $bytesWritten === strlen($output) && fflush($file);
    flock($file, LOCK_UN);
    fclose($file);

    return $written ? 'saved' : 'error';
}

if (($_SERVER['REQUEST_METHOD'] ?? '') !== 'POST') {
    header('Allow: POST');
    showResult('Submission not accepted', 'Please submit the registration form using
the POST method.', [], 405);
    exit;
}

$name = preg_replace('/\s+/u', ' ', postValue('name'));
$email = postValue('email');
$mobile = postValue('mobile');
$course = postValue('course');
$year = postValue('year');
$password = postValue('password');
$confirmPassword = postValue('confirmPassword');
$gender = postValue('gender');
$errors = [];

if ($name === null || !preg_match('/^(?=.{2,60}$)[\p{L}\p{M}]+(?:[ .\'-
][\p{L}\p{M}]+)*$/u', $name)) {
    $errors[] = 'Enter a name with 2–60 letters. Spaces, apostrophes, periods, and
hyphens are allowed.';
}
if (strlen($email) > 254 || !filter_var($email, FILTER_VALIDATE_EMAIL)) {
    $errors[] = 'Enter a valid email address.';
}
if (
    !preg_match('/^\+?[0-9][0-9\s()-]*$/', $mobile)
    || strlen(preg_replace('/\D/', '', $mobile)) < 8
    || strlen(preg_replace('/\D/', '', $mobile)) > 15
) {
    $errors[] = 'Enter a valid mobile number with 8–15 digits.';
```

```
}
if (!in_array($course, [
    'B.Tech Computer Engineering',
    'B.Tech Information Technology',
    'Bachelor of Computer Applications'
], true)) {
    $errors[] = 'Select a valid course.';
}
if (!in_array($year, ['Year 1', 'Year 2', 'Year 3', 'Year 4'], true)) {
    $errors[] = 'Select a valid year of study.';
}
if (!preg_match('/^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[^A-Za-z0-9\s])\S{8,64}$/',
$password)) {
    $errors[] = 'Use 8–64 characters with uppercase and lowercase letters, a number,
and a symbol.';
}
if ($confirmPassword === '' || $confirmPassword !== $password) {
    $errors[] = 'The passwords do not match.';
}
if (!in_array($gender, ['Female', 'Male', 'Non-binary', 'Prefer not to say'], true)) {
    $errors[] = 'Select a valid gender option.';
}
if (!isset($_POST['terms']) || !is_string($_POST['terms']) || $_POST['terms'] !==
'on') {
    $errors[] = 'Accept the terms and conditions to register.';
}

if ($errors) {
    showResult('Please check your details', 'The registration could not be completed.
Correct these items and try again:', $errors, 422);
    exit;
}

$registration = [
    'name' => $name,
    'email' => $email,
    'mobile' => $mobile,
    'course' => $course,
    'year' => $year,
    'gender' => $gender,
    'passwordHash' => password_hash($password, PASSWORD_DEFAULT),
    'submittedAt' => date(DATE_ATOM)
];

$saveResult = saveRegistration($registration);
if ($saveResult === 'duplicate') {
    showResult('Email already registered', 'There is already a registration for this
email address.', [], 409);
```

```
    exit;
}
if ($saveResult !== 'saved') {
    showResult('Could not save registration', 'The registration file could not be
written. Check that the server can write to the studentHUB/data folder.', [], 500);
    exit;
}

showResult('Registration successful', 'Thanks, ' . $name . '! Your registration has
been saved.');
```

5. Result

The PHP form was implemented to accept user input through POST, validate and sanitize submitted values, save valid data to a CSV or JSON file, and display suitable success or validation-error messages.

6. Conclusion

This practical demonstrates server-side form handling using PHP. Validation helps reject missing or invalid values, sanitization supports safer input processing, and file storage provides a simple way to retain submitted records.

— End of Practical 7 Submission —